COMM 291C

Analysis of Skyscrapers (Higher than 100m) in the United States

Jackson Lin

Student ID: 10143326

November 29, 2025

1 Introduction

"In New York
Concrete jungle where dreams are made of
There's nothing you can't do"

— *JAY-Z and Alicia Keys, Empire State of Mind* [1]

Figure 1.1: A shot of the New York City skyline at night, illustrating the dense concentration of skyscrapers.

New York, a city of dreams, is a distinctive symbol of skyline teeming with skyscrapers. Whenever the sun goes down, the dazzling lights from the skyscraper jungle always keep the city youthful and energetic, as if one were watching story after story unfold. Besides the skyscraper's spiritual image, it is usually defined as a tall and habitable building, often at least 100 meters (330ft) in height, for various uses. [2] Although the style and design of skyscrapers inherently vary across the region driven by the geography and cultural context, there still exists something interesting that can be explored: How has the development of skyscrapers in the United States[1] changed over time? Is there a significant relationship between construction materials and building purpose? How are skyscrapers geographically distributed across the country? Driven by these questions in my mind, I selected a secondary dataset found in *StatCrunch* [3] with regard to 2105 skyscrapers completed from 1884 to 2014, and hope to uncover some interesting patterns by analyzing the distribution of skyscrapers with all the knowledge learned in COMM 291C.

[1]Particularly pointing out, this report mainly focuses on skyscrapers in the United States. Since it's a nation leading the world in terms of economic power and urban development from the last century, it is interesting to explore how skyscrapers have evolved over time and space within this country.

2 Dataset Description

First of all, we need to classify variables by type and their measurement scales. Relevant variables are summarized as follows:

Categorical Data:

- **Nominal Scale**
  - Building Name
  - Primary Use
  - Material
  - Location (City, State)

Numerical Data:

- **Ratio Scale**
  - Rank
  - Height (m)
  - Floors
- **Interval Scale**
  - Completed Year

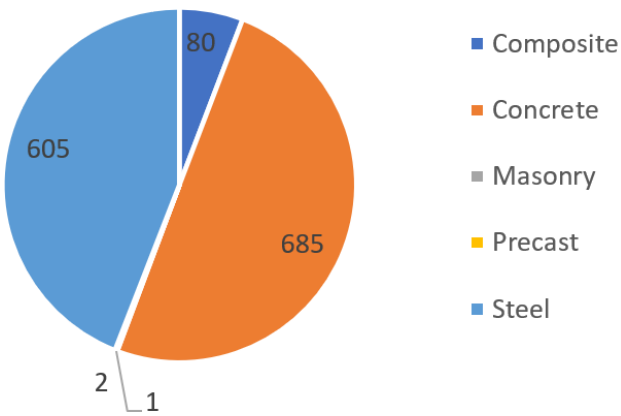
In terms of categorical variables, *Building Name*, *Primary Use*, *Material*, and *Location* are measured on a nominal scale, as their category values express no order or ranking.

On the numerical side, *Rank*, *Height*, and *Floors* are ratio variables because they have a true zero point and allow for meaningful comparisons using multiplication and division. Conversely, *Completed Year* is treated as an interval variable since dividing years is not meaningful (e.g., 2000 is not twice 1000), although differences between years are meaningful.

3 Categorical Analysis

3.1 Building Materials

The building materials are categorized as six groups: Concrete, Steel, Composite, Masonry, Precast and an ”Others” group. Particularly, ”Others”[2] refers to buildings with missing material information in the original dataset. For better analysis, this group was removed.



Material	Count	Percentage
Composite	80	5.83%
Concrete	685	49.89%
Masonry	1	0.07%
Precast	2	0.15%
Steel	605	44.06%
Grand Total	**1373**	**100.00%**

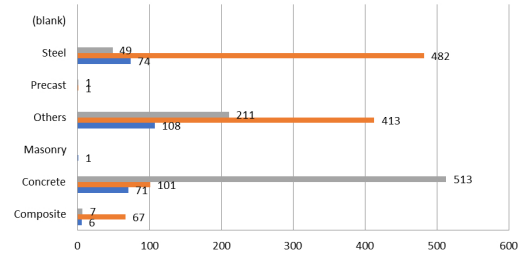
Figure 3.1: Left: Pie chart of distribution of construction materials; Right: Frequency and percentage table of construction materials.

The dataset shows that the construction of skyscrapers in the United States mainly relies on concrete and steel as “primary materials”, which account for 49.89% and 44.06% of the skyscrapers respectively. This result is consistent with Elnimeiri and Hwang’s finding[4], which suggests the original dataset has accountability to some extent. By contrast, composite(5.83%) and less common materials, precast(0.15%) and masonry(0.07%), show far less frequency in the dataset.

Since it is somewhat monotonous to solely see the distribution of material, it occurred to me that maybe there exists a relationship between construction materials and primary usage. However, for better analysis, groups of ”Masonry” and ”Precast” are excluded for their triviality (too negligible). The relevant clustered bar chart was created in Excel and is shown in Figure 3.2 on the next page.

[2]This implies that the author was practicing a sort of ”minimalism” when collecting data, as he left 34.77% of the entries without recorded material information. (just joking)

3.2 Materials vs Use



	Composite	Concrete	Masonry	Others	Precast	Steel	(blank)
(blank)							
Residential	7	513		211	1	49	
Office	67	101		413	1	482	
Mixed	6	71	1	108		74	

$\Longrightarrow$

Use (Primary)	Composite	Concrete	Steel
Residential	7	**513**	49
Office	67	101	**482**
Mixed	6	71	74

Figure 3.2: Left: Clustered bar chart of primary use by material ; Right: Frequency and percentage table of material categories.

From the clustered bar chart above, it is notable that concrete is largely used in residential skyscrapers, with a significant count of 513 buildings. In contrast, steel rules office towers, a substantial count of 482 buildings. Mixed-use skyscrapers see a more balanced distribution between concrete and steel, with counts of 71 and 74 respectively. Composite materials are used less frequently across all categories, with office buildings having the highest count of 67.

But why does such a pattern exist? One possible explanation comes from the material properties themselves. Concrete is cheaper compared with steel and has strong ”durability” [5], which means it can withstand things like blazes and earthquakes. Since residential towers are usually less profitable than office buildings, developers prefer concrete because it keeps the cost low while still providing enough comfort and safety for people to live in. On the other hand, steel can be installed much more quickly [5] and it is easier to transport or recycle. Although the cost is higher, these advantages tend to outweigh the expense. For corporations, especially those seeking to stay ahead of competitors, faster construction means they can open new branches and expand their operations more efficiently. This makes steel a practical choice for office skyscrapers.

Besides, since we have already created a 3×3 contingency table with regard to materials and use, it would be a waste not to run a full *independence test* to formally examine whether there is a significant association between primary use and construction material. Now, we have the relevant table on the next page again.

3.3 Independence test

Use (Primary)	Composite	Concrete	Steel	Total
Residential	7	513	49	569
Office	67	101	482	650
Mixed	6	71	74	151
Total	80	685	605	1370

Table 3.1: Contingency table of building use and material type

In this section, probability works. Let U_i denote the i-th Use category and M_j the j-th Material category (e.g., U_2 = "Office"). Each cell corresponds to the joint event $U_i \cap M_j$.[3] By definition, $P(U_i|M_j) = \frac{P(U_i \cap M_j)}{P(M_j)}$. In that case, if U_i and M_j are independent, which means under the condition of M_j, the probability of U_i is still itself, so $P(U_i|M_j) = P(U_i)$, which implies $P(U_i \cap M_j) = P(U_i)\,P(M_j)$.

To be more specific, we choose one cell as an example. Let's say the cell of residential and composite. Then, $P(U_1 \cap M_1) = \frac{7}{1370} \approx 0.0051$. and $P(M_1) = \frac{80}{1370} \approx 0.0584$. After that, the conditional probability of residential under composite is $P(U_1|M_1) = \frac{P(U_1 \cap M_1)}{P(M_1)} = 0.0875$. Since $P(U_1) = \frac{569}{1370} \approx 0.4153$, which is not equal to $P(U_1|M_1) = 0.0875$, we can conclude that residential and composite are not independent. Similarly, we can check other cells one by one, and find that none of them are independent. However, this is not a formal test yet. To formally examine whether there is a significant relationship between primary use and construction material, we can perform a Chi-square Test of Independence.

[3] I absolutely respect different notations in different contexts, so I am just pointing out that $P(A \text{ and } B) = P(A) \cdot P(B)$.

3.4 Going Further: Chi-square Test of Independence

The reason why the previous method is not formal is that it only checks whether two events are independent, which is not enough to conclude the whole table. However, the Chi-square Test of Independence helps us to examine the relationship between two categorical variables in a contingency table as a whole.

- H_0: There is no relationship between primary use and construction material of skyscrapers in the U.S.A.
- H_1: There is a relationship between primary use and construction material of skyscrapers in the U.S.A.

Then, to test the null hypothesis H_0, we need to calculate the Chi-square test statistic by using the formula below:

$$\chi^2_{STAT} = \sum_{\text{all cells}} \frac{(f_o - f_e)^2}{f_e}$$

Take the cell of residential and composite as an example again. The observed frequency is $f_o = 7$. To calculate the expected frequency f_e, we use the multiplication rule for independent events. Basically, $f_e = P(A \cap B) \cdot n$, where n is the total number of observations (1370 in this case). Thus, using the probability we computed earlier, we have $f_e = P(U_1 \cap M_1) \cdot n = P(U_1) \cdot P(M_1) \cdot 1370 = \frac{569}{1370} \cdot \frac{80}{1370} \cdot 1370 \approx 33.23$.

Later on, we compute the expected frequencies for each cell in the contingency table and write it as a form of matrix.

$$O = \begin{pmatrix} 7 & 513 & 49 \\ 67 & 101 & 482 \\ 6 & 71 & 74 \end{pmatrix} \qquad E = \begin{pmatrix} 33.23 & 284.50 & 251.27 \\ 37.96 & 325.00 & 287.04 \\ 8.82 & 75.50 & 66.68 \end{pmatrix}$$

Then, $df = (c-1)(r-1) = (3-1)(3-1) = 4$, and we choose $\alpha = 0.05$. All set, then we can calculate the Chi-square test statistic using the formula, $\chi^2 = \sum \frac{(O-E)^2}{E} = \sum_{i=1}^{3} \sum_{j=1}^{3} \frac{(O_{ij}-E_{ij})^2}{E_{ij}} \approx 678.05 > \chi^2_{.050} = 9.488$. [6] In that case, since χ^2_{stat} falls in the rejection region, we reject the null hypothesis, which indicates that there is a significant relationship between primary use and construction material of skyscrapers in the U.S.A.

3.5 Geographic Distribution

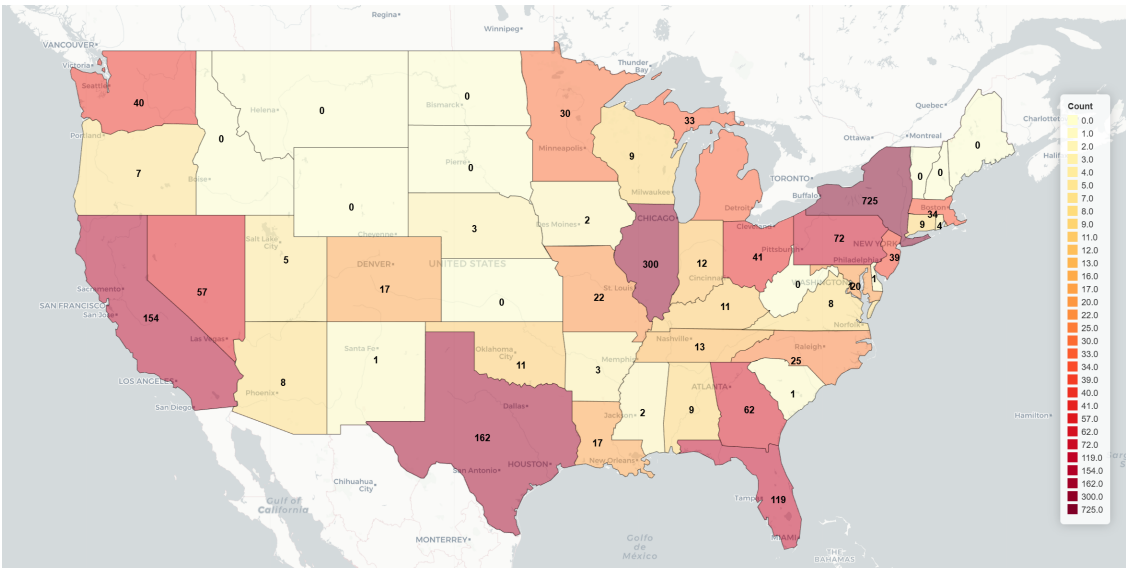


Figure 3.3: Geographic distribution of skyscrapers in the contiguous United States (Interactive Map See [7])

Generally, skyscrapers are unevenly distributed across the United States. When catching the first glimpse of the map[4], the readers' eyes are instantly drawn to the conspicuous "pig-liver red" regions: New York on the East Coast, Illinois beside Lake Michigan, Texas in the South, California on the West Coast, and Florida, the slender peninsula jutting confidently into the ocean. Among them, New York ranks the highest with an astonishing 725 skyscrapers, even more than double the second place, Illinois (300). The remaining three states, Texas, California, and Florida are relatively evenly matched, with counts 162, 154, and 119 respectively. However, they also show a similar pattern: along the waterfront areas, especially the coastlines and lakeshore. This makes intuitive sense since these locations often serve as major economic and cultural hubs, which can attract residents and entrepreneurs; the demand for vertical expansion naturally follows.

Apart from these prominent skyscraper states, several regions show a moderate presence of skyscrapers, including Washington D.C. (61), Georgia (36), and Massachusetts (46), which form

[4]As the locations of skyscrapers are given in the form of city and state names, for better visualization, my original thought was to generate a geographic distribution map just by using Excel. However, the relevant feature is unavailable probably due to some sensitive and controversial political issues. As a result, I turned to Python programming in Jupyter Notebook, utilizing GeoPandas and Folium libraries to create interactive maps by following a similar idea from Dylan Song. [8]

noticeable clusters along the East Coast, while Arizona (31) and Nevada (40) stand out in the Southwest, and Colorado (22) appears as the primary high-rise center in the Mountain West. In contrast, much of the Midwest and interior West contains only a handful of skyscrapers: Montana, Wyoming, and the Dakotas each register fewer than five, with some showing only one or two. These clear numerical gaps highlight the sharp regional imbalance in urban development and economic activity across the United States.

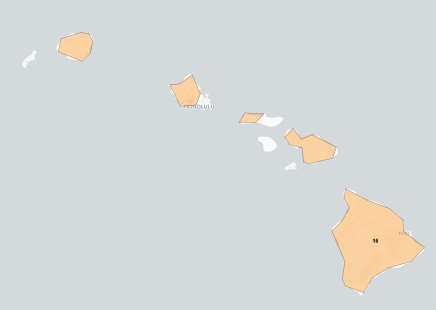


Figure 3.4: Skyscrapers in Hawaii (Interactive Map See [7])

Figure 3.5: Skyscrapers in Alaska (Interactive Map See [7])

At the national edges, Alaska and Hawaii diverge sharply: Alaska has no skyscrapers at all, while Hawaii shows a small, uneven cluster—most notably 16 on the Big Island. These differences reflect how geography and population density shape vertical development.

4 Numerical Analysis

4.1 Numerical Descriptive Statistics

To summarize the central tendency, variability and shape of the numerical variables (height, number of floors), I used Excel to compute the descriptive statistics shown in Table 4.1.

Table 4.1: Summary table of key numerical variables.

Variable	Min	Q_1	Median	Q_3	Max	Range	Mean	STD	CV
Height (m)	100	115.3	132.6	159.7	541.3	441.3	143.70	41.26	0.287
Floors	13	30	36	43	110	97	37.54	10.62	0.283

To begin with, we examine height and floors, as they appear first in the summary table above. The highest skyscraper is the rebuilt One World Trade Center with about 541.3m (94 floors). [9] Correspondingly, the skyscraper with the most floors is the former one[5] , with 110 floors. (417 m) Pivoting to the lower end of the distribution, the Civil Courts Building has the fewest floors (13 floors at 117.7 m), while the East River Tower Apartments represent the minimum height category (100 m with 33 floors). These findings not only reflect the wide range of skyscrapers in terms of height and floors but also indicates the potential inconsistency of floor-to-height ratios across different buildings.

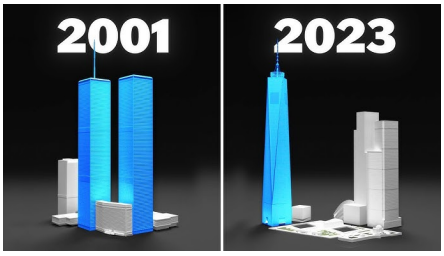


Figure 4.1: The former World Trade Center towers (with antenna) (2001) and the current One World Trade Center (2023).

Now turning to central tendency, the mean height is about 143.7 m, while the median height is 132.6 m, which is relatively close. The mean number of floors is about 37.54, with a median

[5] While checking the data, I noticed something unusual: there were two entries named "One World Trade Center," but their heights and floor counts didn't match. After looking it up, I found that the dataset was referring to two different buildings from different time periods, which explains why the numbers don't line up.

of 36 floors. In both cases, the mean is greater than the median, which implies a right-skewed distribution.

Regarding variation, the standard deviation (STD) for height is 41.26 m, and for floors, it is 10.62 floors. The coefficient of variation (CV) for height is approximately 0.287, and for floors, it is about 0.283. These CV values indicate a moderate level of relative variability in both height and floor counts among the skyscrapers in the dataset.

To reveal the skewness of their distribution, boxplots were created below, since they rely on resistant measures such as the median and quartiles. In other words, even though the dataset contains a few extremely tall buildings (e.g. $Z_{\max} = \dfrac{H_{\max} - \bar{H}}{s} = \frac{541.3-143.7}{41.26} = 9.636 > 3 \rightarrow$ Outlier), median and quartiles hardly change. Boxplots below indicate that both of them have right-skewed distributions, as the upper whiskers extend further than the lower whiskers.

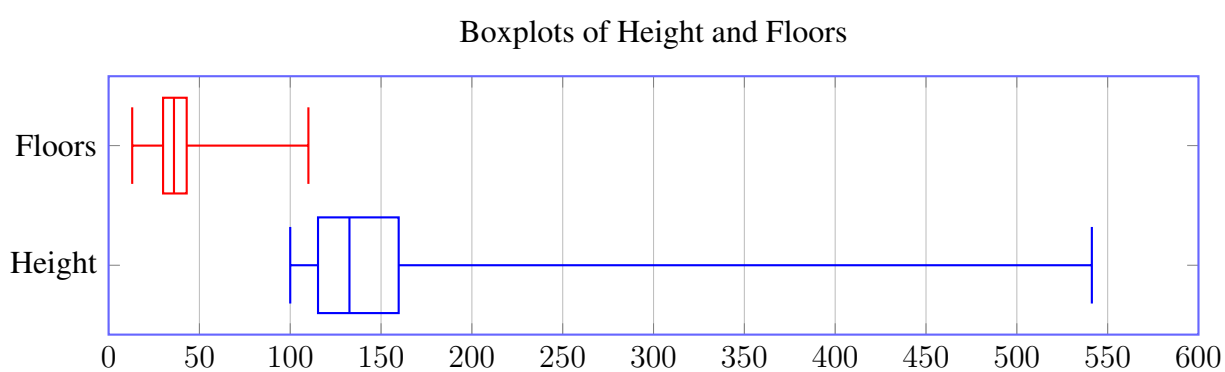


Figure 4.2: Boxplots of Height and Floors

4.2 Correlation

In the previous section, we have already observed that height and floors share similar distributional characteristics: same right-skewness, close central tendency and variation measures. Also, just by common sense, more floors usually means taller buildings. Thus, we calculated covariance ($COV \approx 359.312$) and coefficient of correlation ($CORR \approx 0.828$) between them and constructed a scatter plot in Excel to see the relationship between height and floors.

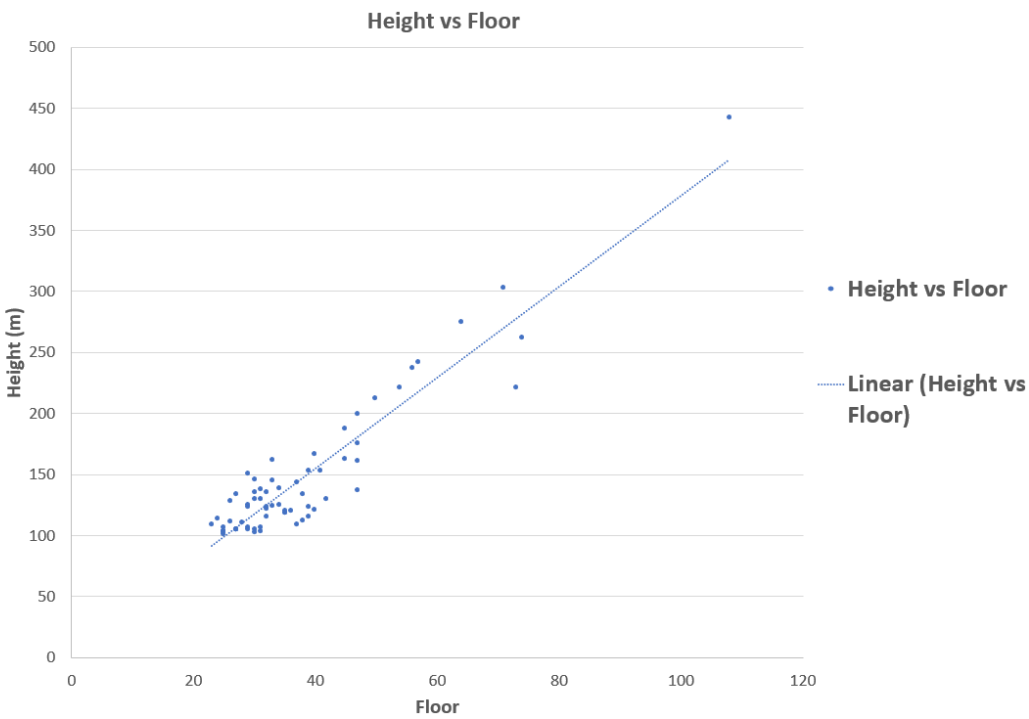


Figure 4.3: Scatter plot of Height vs Floors with trendline.

Since COV is positive and $CORR$ is close to 1, it reflects a strong positive linear relationship between height and floors. Meanwhile, the scatter plot supports it further, as the points cluster closely around the upward trend line, especially the region of lower heights and fewer floors. Our plot also supports and demonstrates the right-skewed distribution we saw before. Overall, the result suggests that the height of the skyscraper tends to increase with the increase of floors, confirming our prior intuition about their relationship. However, it is worth noting that the correlation does not mean cause and effect. Other factors, such as the design and purpose, can affect the ratio of height to floors in skyscrapers. Therefore, their linear relationship is strong but not absolute.

4.3 Trends in Construction Periods

Since the total count of skyscrapers is 2105, which is somewhat too large to construct a time-series plot. By grouping the completed years into 10-year periods, we can better visualize the trends in skyscraper construction over time.

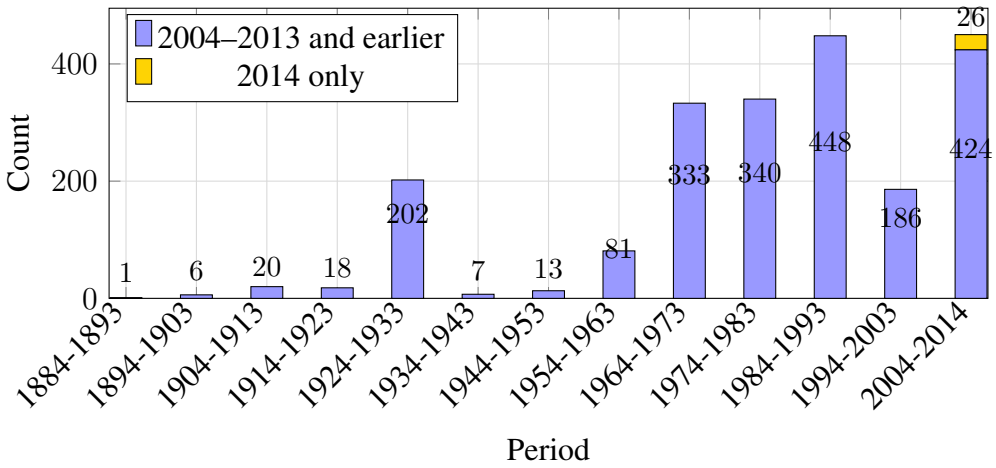


Figure 4.4: Number of skyscrapers completed by period (2014 separated within 2004–2014).

Overall, there were several construction waves during the whole time span. As shown in Figure 4.4, the number of skyscrapers completed remained relatively low in the first seven decades (1884-1953), with counts generally below 20 per decade, except for a notable spike in the 1924-1933 period, which saw 202 skyscrapers completed. After 1954, the pace of construction began to accelerate, with a clear upward trajectory through the post-war decades. Activity peaked in the 1984–1993 period with 448 completed skyscrapers, followed by a plummet of 262 skyscrapers between 1994 and 2003. The trend continued into the early twenty-first century, culminating in 424 completions during 2004–2013, along with an additional 26 buildings completed in 2014 alone.

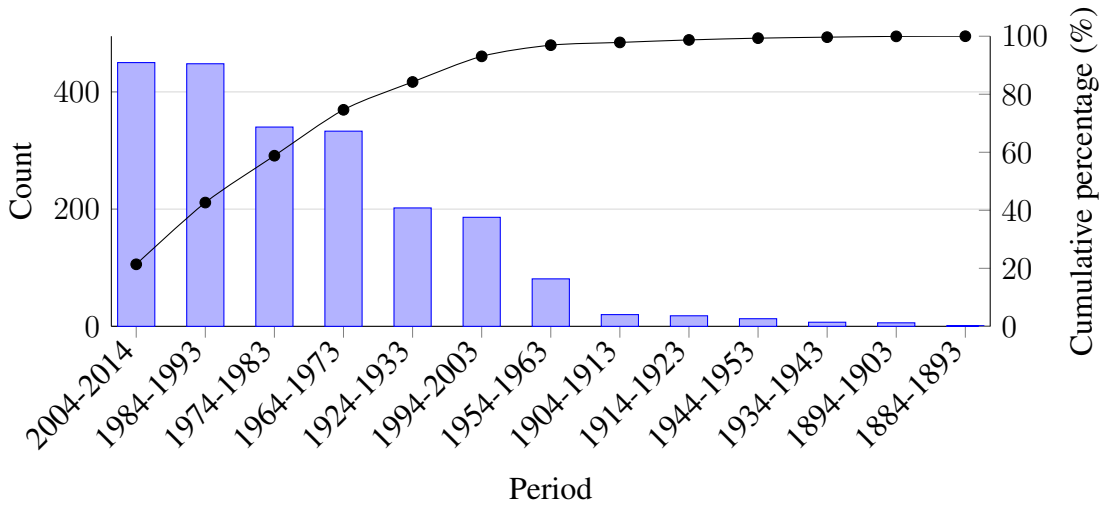


Figure 4.5: Pareto chart of skyscraper completion periods (sorted by frequency).

Also, to identify which periods contributed most substantially to the overall skyscraper stock, a Pareto chart was constructed. The chart shows that just four periods are the **"Vital Few"**, including 1964–1973, 1974–1983, 1984–1993, and 2004–2013, which account for more than half of all skyscrapers in the dataset. This concentrated pattern implies that the development of U.S. skyscrapers tends to occur in specific construction periods rather than being evenly distributed across the span.

5 Conclusion

From a COMM 291C point of view, this project ended up being a hands-on review of most of what we did in class. I started by cleaning and sorting the skyscraper data into categorical and numerical variables, and then matched each one with a scale that actually made sense in context (nominal for labels such as city and material, ratio for height, floors and rank, and interval for completed year).

On top of that structure, I tried to pick graphs that were not just "pretty" but also tied to the type of variable: pie and clustered bar charts for materials and uses, maps for locations, boxplots for height and floors, and decade-based bar charts plus a Pareto chart to show how construction activity changed over time.

For numerical summaries, I used the standard toolkit from the course — mean, median, quartiles, range, standard deviation, coefficient of variation and Z-scores — to describe the distributions of height and floors, and then used covariance and the correlation coefficient to quantify how closely they move together. For the categorical side, I built a 3×3 contingency table for material by primary use, worked with joint and conditional probabilities to talk about independence, and finally ran a Chi-square Test of Independence as the formal check. Of course, everything here is still descriptive and depends on this particular StatCrunch dataset, which has missing material values and only includes buildings over 100m.

So, in a way, this report is just COMM 291C in miniature: organizing variables, choosing sensible charts, computing the right summary measures and using basic probability ideas to turn a big secondary dataset into a story about how U.S. skyscrapers are built, used and distributed.

All in all, the big picture is clear enough. For the truly loyal readers who made it all the way here, "attention is all you need" does not seem entirely wrong. Now I can finally log off from VSCode and LaTeX, and you can finally stop staring at yet another skyscraper-sized report. Thanks for reading, and take care.

References

[1] JAY-Z, and Alicia Keys. "Empire State of Mind." *YouTube*, uploaded by JAY-Z, 2009. https://youtu.be/vk6014HuxcE?si=JxZUKB2_ADJJydCa.

[2] "Skyscraper, Emporis Standards". *Emporis.com.* Archived from the original on 11 May 2015. https://web.archive.org/web/20150511222640/http://www.emporis.com/building/standard/75/skyscraper.

[3] Cecil College, "Skyscrapers in the U.S." *StatCrunch*, 2015. https://www.statcrunch.com/.

[4] Elnimeiri, Mahjoub M., and Youngjin Hwang. "Towards Sustainable Structure of Tall Buildings by Significantly Reducing the Embodied Carbon." *Sustainability*, vol. 17, no. 6, 2025, article 2754. https://doi.org/10.3390/su17062754.

[5] NY Engineers. "Steel vs Concrete: Which Is Better for Tall Buildings?" *NY Engineers*. https://www.ny-engineers.com/blog/comparing-steel-construction-and-concrete-construction.

[6] "Chi-Square Distribution Table." *MIT OpenCourseWare*. https://ocw.mit.edu/courses/14-30-introduction-to-statistical-method-in-economics-spring-2006/bdbe9805a1722db51474900d77e16438_t_table.pdf.

[7] Pitachin. "US Skyscraper Geographic Distribution Interactive Map." *GitHub*, 9 Nov. 2025. https://pitachin.github.io/US_Skyscrapers_Map/US_Skyscrapers_Explore.html.

[8] Song, Dylan. "How to Create INTERACTIVE MAPS in Python." *YouTube*. https://youtu.be/_xeoUYUO8Lo?si=etnst6TqwVxszS7i.

[9] Wikipedia contributors. "September 11 Attacks." *Wikipedia, The Free Encyclopedia*. https://en.wikipedia.org/wiki/September_11_attacks.